

With SFT and GRPO

Exploring LoRA for
Math Reasoning

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Definition

Intro

Modern LLM training pipeline =

Pre-training + Post-training

Foundation-building Stage

Learn general language ability from large-scale data.

Behavior-shaping & Specialization Stage

Adapt the pretrained model for instruction following, alignment, and specific tasks.

Post-training

Training Methodology

SFT - Supervised Fine-Tuning

on *input-output examples* to teach the model how to respond.

RFT - Reinforcement Fine-Tuning

improves the model using *rewards* instead of only fixed target answers.

└─ **GRPO** - Group Relative Policy Optimization

an RFT method that trains the model by *rewarding better answers* within a group of sampled outputs.

Model Update Methodology

PEFT - Parameter-Efficient Fine-Tuning

updates only a *small subset of parameters* to adapt a pretrained model more cheaply.

└─ **LoRA** - Low-Rank Adaptation

a PEFT method that freezes the base model and learns *small low-rank adapter weights* instead of updating all parameters.

02

Motivation & Literature Review

Motivation -

Paper Learning to Reason in 13 Parameters*

7,000,000,000+

Full model Qwen2.5-7B-Instruct parameters update on GSM8K (accuracy **76%-97%****)

1,000,000+

Standard LoRA / SFT setup update on GSM8K (accuracy **76%-98%**)

13

TinyLoRA / GRPO on GSM8K (accuracy **76%- 91%**)

What is
TinyLoRA ?

* <https://arxiv.org/pdf/2602.04118>

** estimated from the chart in the paper

03

Methodology

Fine-Tuning Lineage

LoRA

2021

1



Low-rank adapters make model adaptation cheap enough to compare multiple ranks locally.

GRPO

2024

2



DeepSeekMath brought GRPO into reasoning RL and motivates group-relative optimization for math.

LoRA-XS

2024

3



Uses frozen SVD factors plus a tiny trainable core to cut parameters drastically.

TinyLoRA

2026

4



13 Parameters pushes the update budget toward only a handful of trainable parameters.

Concepts

LoRA

Keeps the base model frozen and learns a **low-rank update** through small adapter weights.

LoRA-XS

Builds on LoRA by using **fixed SVD directions** from the base weights and training only a much smaller core.

TinyLoRA

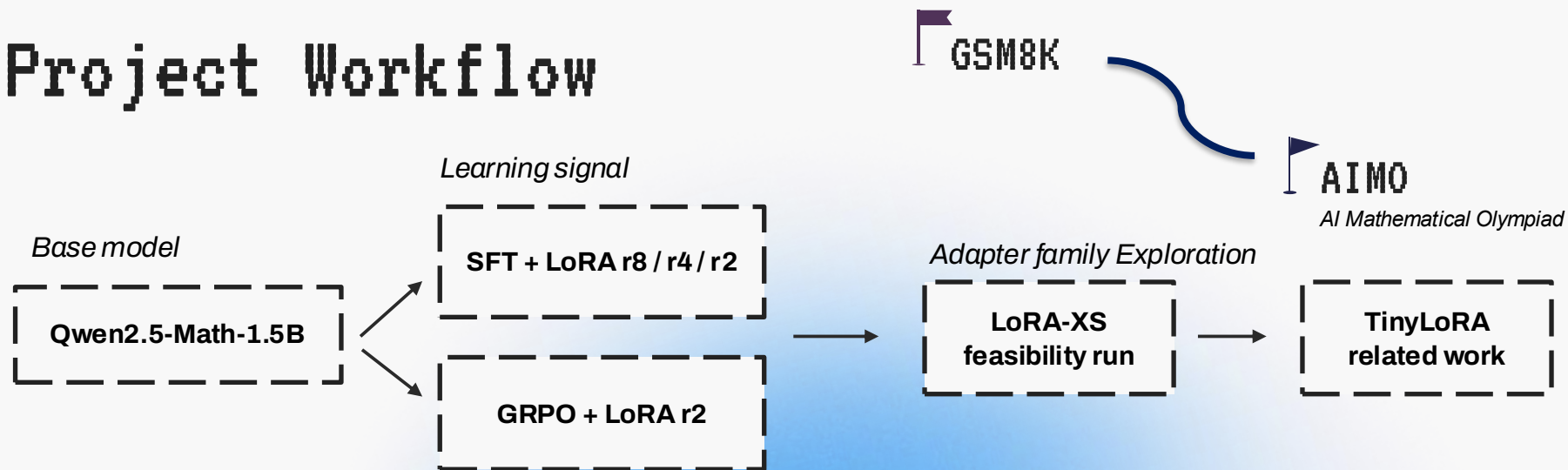
Pushes parameter reduction even further by replacing the trainable core with a **tiny projected trainable vector**.

Method	Main idea	Pros	Cons
LoRA	Low-rank adapters	Simple, effective, widely used	Still needs more parameters than newer variants
LoRA-XS	Fixed SVD structure + small core	Smaller and more parameter-efficient than LoRA	More specialized, less standard
TinyLoRA	Tiny projected trainable vector	Extreme parameter efficiency	More experimental, harder to validate broadly

04

Project Workflow

Project Workflow



- ❑ Inspired by *Learning to Reason in 13 Parameters*, which investigates how far ultra-small RL updates can improve reasoning.
- ❑ Started with **SFT + LoRA** on **Qwen2.5-Math-1.5B** with ranks **8 / 4 / 2** in a local notebook workflow.
- ❑ Now pivoting to **GRPO + LoRA (rank 2)**, moving from supervised targets to reward-based optimization.
- ❑ Evaluate on **GSM8K** and **Olympiad-style math** to test whether reasoning improvements transfer to harder problems.
- ❑ Include **LoRA-XS** as a feasibility check; keep **TinyLoRA** as related work and future work.

05

Results Review

Results at a Glance

Notebook run on Reasoning (GSM8K) -100 problems, Olympiad-100 problems, and AIMO-150 problems

Reasoning-100

SFT LoRA r=2
ACC = 0.88

Highest score on Reasoning-100; about +0.02 over the base model.

Olympiad-100

SFT LoRA
r=2 / r=4

Top Olympiad result; both r=2 and r=4 reached 0.59.

AIMO-150

SFT LoRA r=4
0.733

Best AIMO score in the final run; the harder set favored a slightly larger adapter.

GRPO LoRA r=2 params

272K

Smallest trainable budget among the compared methods, about 8.5× smaller than SFT LoRA r=2.

SFT LoRA r=2 params

2.3M

Still modest relative to full fine-tuning, but much stronger in raw accuracy across the three sets.

GRPO runtime

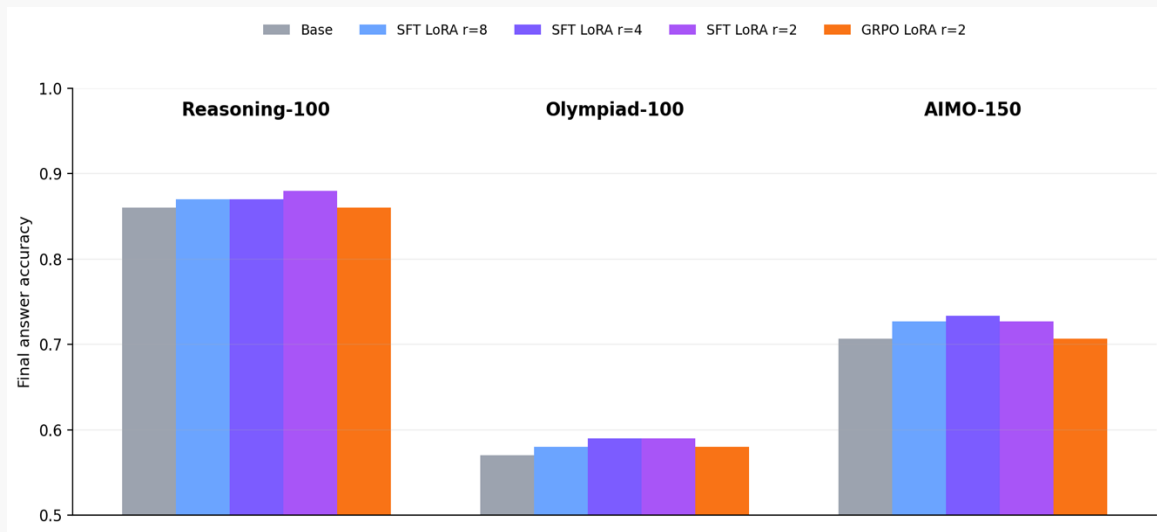
30,838.75s

Much slower than the SFT runs in this CPU setup, even though its adapter was smaller.

SFT + LoRA has better accuracy, while GRPO remained the most compact trainable setup.

Benchmark Performance

Accuracy across all evaluated conditions



Reasoning-100

Base was already strong at 0.86, but SFT LoRA r=2 still improved it to 0.88. GRPO matched the base and did not add extra gain in this run.

Olympiad-100

This was the closest benchmark: SFT LoRA r=2 and r=4 both reached 0.59, while GRPO reached 0.58. The gains were modest but consistent.

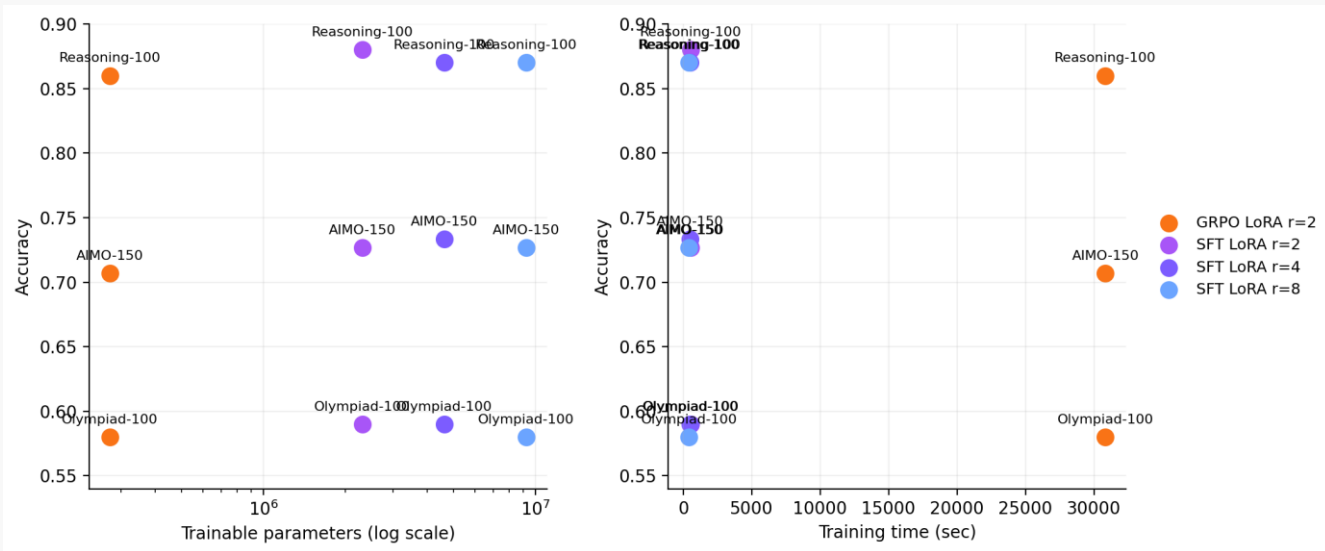
AIMO-150

AIMO favored a slightly larger adapter: SFT LoRA r=4 achieved the best score at 0.733, outperforming both r=2/r=8 and GRPO.

Pattern: Smaller LoRA ranks were enough to help, but the best rank depended on benchmark difficulty and distribution.

Efficiency Trade-offs

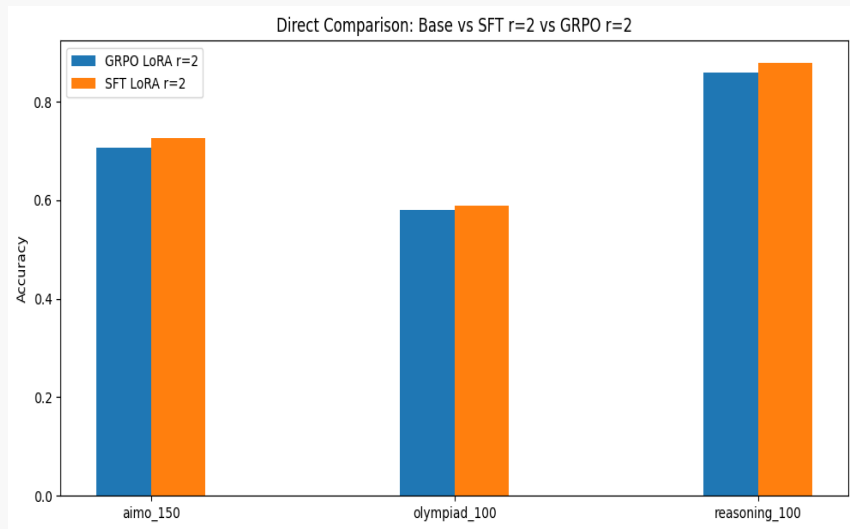
Parameter budget, training time, and where GRPO was strongest



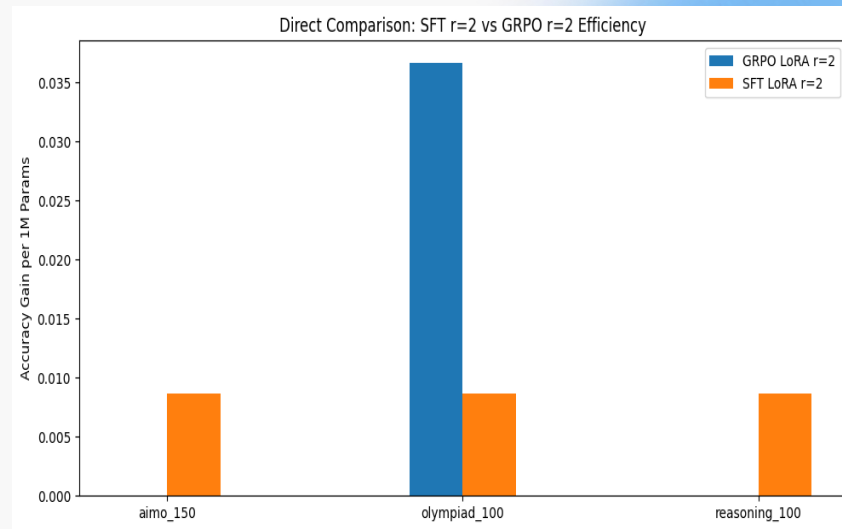
- GRPO LoRA r=2 had the smallest trainable budget: 272k parameters (0.0176% of the model), versus 2.31M / 4.62M / 9.23M for SFT r=2 / r=4 / r=8.
- That parameter efficiency did not translate into time efficiency here: GRPO took about 30,839 seconds to train, while the SFT runs finished in about 439–555 seconds.
- SFT LoRA r=8 was the fastest trained model, yet it was not the most accurate; in practice, r=2 or r=4 gave a better accuracy-cost balance.

SFT VS GRPO with LoRA Rank=2

Same low-rank budget, different training objectives



Accuracy: Base vs SFT r=2 vs GRPO r=2

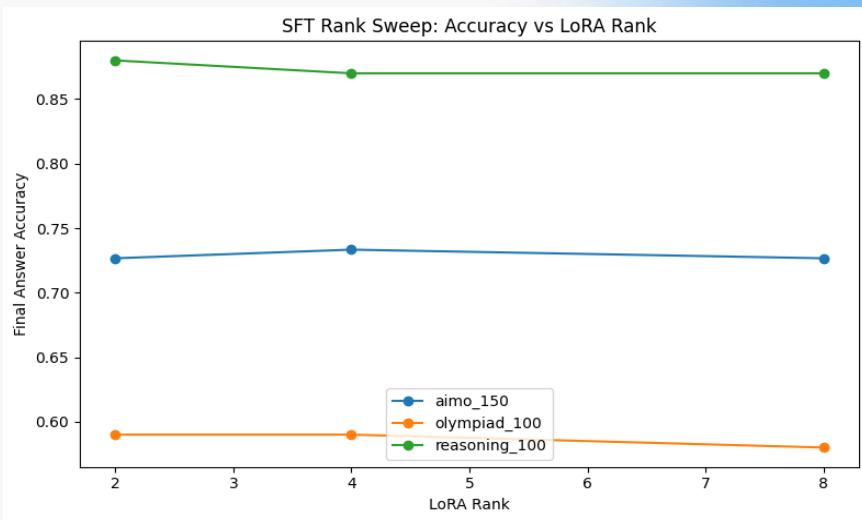


Efficiency: accuracy gain per 1M trainable parameters

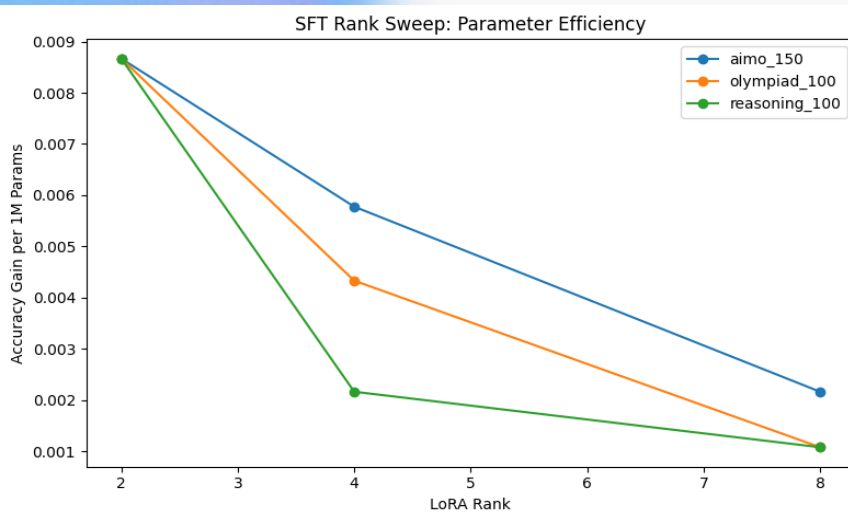
- SFT r=2 beat GRPO r=2 on all three benchmarks in raw accuracy: 0.88 vs 0.86 on Reasoning-100, 0.59 vs 0.58 on Olympiad-100, and 0.727 vs 0.707 on AIMO-150.
- GRPO r=2 was much more parameter-efficient on Olympiad-100: 0.0367 accuracy gain per 1M trainable parameters vs 0.0087 for SFT r=2.

SFT Rank Sweep

Smaller adapters remained strong, but the best rank still depended on the benchmark

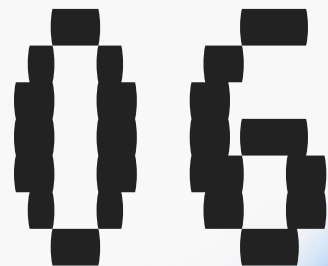


Final answer accuracy vs LoRA rank



Accuracy gain per 1M parameters vs LoRA rank

- Rank 2 was best on Reasoning-100 (0.88), showing that a very small adapter was enough to improve the strongest starting benchmark.
- Rank 4 was best on AIMO-150 (0.733), while Olympiad-100 tied at 0.59 for $r=2$ and $r=4$.
- Parameter efficiency dropped as rank increased, larger adapters added cost faster than they added accuracy.



Key Learnings

Key Learnings

- Small LoRA adapters were enough to move reasoning quality. Even rank 2 improved the base model on Reasoning-100 and tied for best on Olympiad-100.
- The best LoRA rank was benchmark-dependent: rank 2 fit the shorter reasoning tasks best, while rank 4 was strongest on the harder AIMO-150 set.
- In this local notebook workflow, GRPO looked more promising as a parameter-efficiency experiment than as a raw-performance winner. It showed the smallest update budget, but not the best scores

Limitations

- This project is a compact local-environment experiment, not a definitive benchmark of post-training methods.
- The comparison is informative but narrow: SFT + LoRA was tested at ranks 8 / 4 / 2, while GRPO + LoRA was tested only at rank 2.
- Held-out evaluation covered Reasoning-100, Olympiad-100, and AIMO-150; larger benchmark slices would make the conclusions more robust.
- The shared deterministic evaluation protocol improved fairness, but training conditions were not perfectly matched.
 - SFT + LoRA: local MLX workflow with a more efficient training path
 - GRPO + LoRA: local CPU workflow using Transformers + TRL + PEFT

GitHub : <https://github.com/Timidalucia/Mini-Research-Problem>

YouTube : <https://www.youtube.com/watch?v=1Y2c67uLCPs>

Reference :

Morris, J. X., Mireshghallah, N., Ibrahim, M., & Mahloujifar, S. (2026). Learning to Reason in 13 Parameters. ArXiv. <https://arxiv.org/abs/2602.04118>

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Bałazy, K., Banaei, M., Aberer, K., & Tabor, J. (2025). LoRA-XS: Low-Rank Adaptation with Extremely Small Number of Parameters. ArXiv. <https://arxiv.org/abs/2405.17604>

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THANK YOU